

IMAGE CLASSIFICATION USING DEEP LEARNING

Project Report

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1. Abstract

Image classification is a fundamental task in computer vision that enables computers to identify and categorize images automatically. This project presents a web-based Cat and Dog Image Classification system developed using TensorFlow, Keras, MobileNetV2, and Flask. The system uses transfer learning to classify uploaded images into either "Cat" or "Dog". A modern web interface built with Flask allows users to upload an image and instantly receive the prediction along with the confidence score. The project demonstrates the practical application of deep learning in image recognition while maintaining high accuracy and fast inference.

2. Problem Statement

Develop a Cat and Dog Image Classification Web Application using Deep Learning (MobileNet) and deploy it using a Flask API.

3. Tech Stack

- Python
- TensorFlow & Keras
- MobileNetV2
- Flask
- NumPy
- Pillow (PIL)

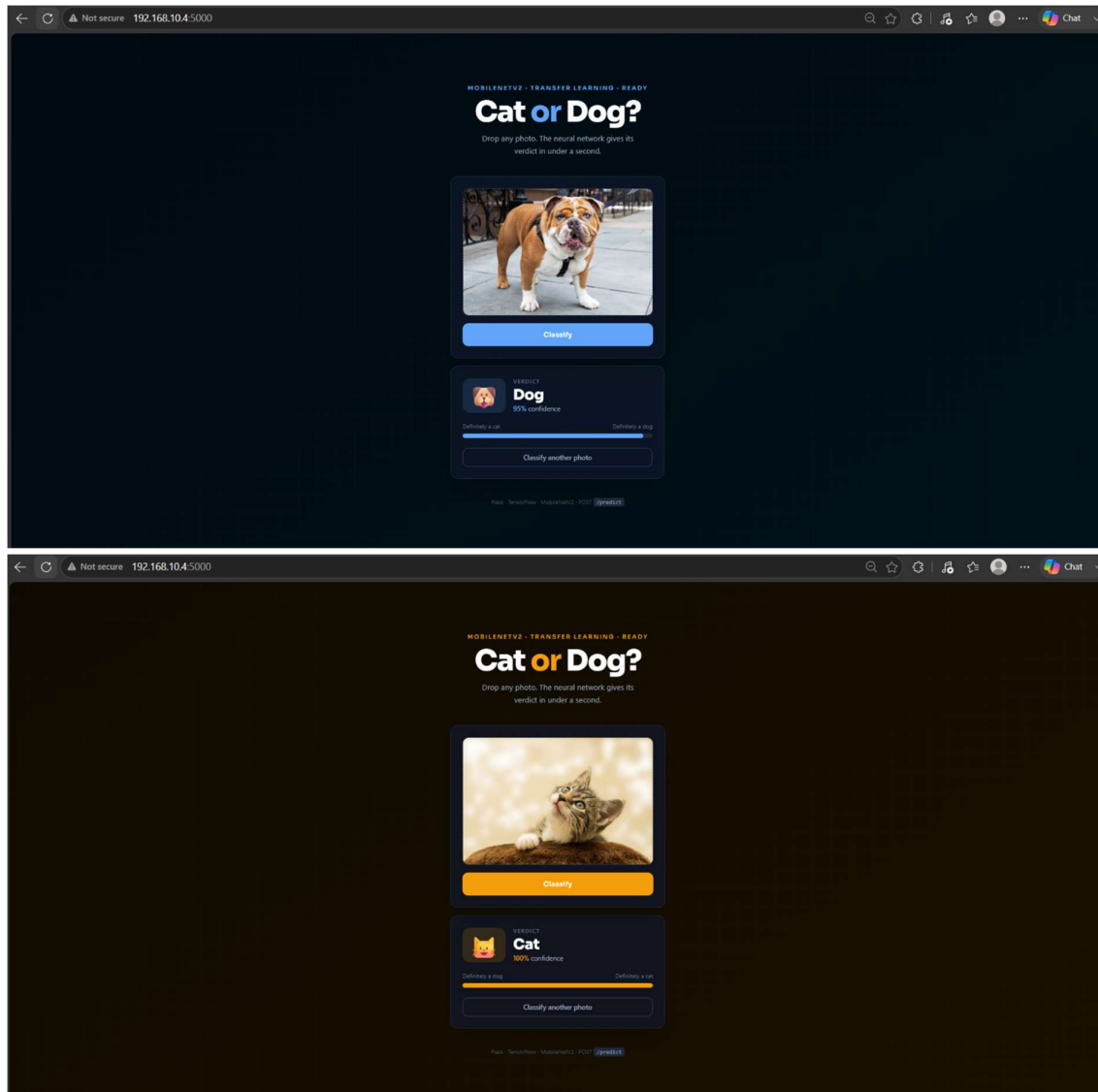
4. Features

- Image upload and preview
- Cat/Dog classification
- Confidence score display
- Responsive and modern user interface

5. Results

The application successfully classifies uploaded images into Cat or Dog with high accuracy and provides fast predictions using the trained MobileNetV2 model.

6. Gallery



7. Conclusion

The project demonstrates how transfer learning can be used to build an accurate and efficient image classification system. By integrating TensorFlow with Flask, the model is deployed as an easy-to-use web application that performs real-time predictions.

8. GitHub Repository

Repository: <https://github.com/nandhanamithra/IMAGE-CLASSIFICATION>